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Psychological safety as a key moderator: exploring digital leadership, flow and improvisational ability through a dual PLS-SEM and fsQCA approach

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Abstract
This study examines how digital leadership influences employees' cognitive and affective competencies, emphasizing the mediating role of flow at work and the moderating effect of psychological safety. Drawing on flow theory, it investigates how digital leadership enhances improvisational ability (cognitive dimension) and emotional commitment (affective dimension) in knowledge-based industries undergoing digital transformation. Adopting a mixed-method analytical approach, the study employs partial least squares structural equation modeling (PLS-SEM) to test hypotheses. The fuzzy-set qualitative comparative analysis (fsQCA) used to uncover multiple pathways leading to high improvisational ability and emotional commitment. Data were collected through three-wave survey with each wave administered one month apart. Total sample comprised of 405 respondents from knowledge intensive sectors. The PLS-SEM results reveal that digital leadership positively affects flow at work, which in turn significantly enhances both improvisational ability and emotional commitment. Flow at work significantly mediates the relationship between digital leadership and employees' improvisational ability and emotional commitment. Moreover, psychological safety strengthens the link between flow at work and employee outcomes. The fsQCA findings identify flow at work and psychological safety as core conditions for achieving high improvisational ability and emotional commitment. This study advances theoretical understanding of digital leadership by identifying flow at work as a critical mechanism and psychological safety as a boundary condition in fostering employee competencies, offering practical insights for organizations navigating digital transformation.

Keywords Digital leadership, Flow at work, Psychological safety, Improvisational ability and emotional commitment

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Introduction

Digital transformation is reshaping how organizations coordinate work, interact with customers and generate value [33]. As firms adopt advanced digital technologies, the role of digital leadership has become increasingly important for guiding employees through new forms of work and heightened uncertainty [25]. Digital leaders support technology adoption, enhance digital literacy and structure work to help employees navigate increased complexity [2]. Although these roles are well acknowledged, research has not fully explained how digital leadership shapes specific employee capabilities that are essential during digital transformation, particularly employees' improvisational ability and emotional commitment.

Improvisational ability is defined as the ability of employees to react creatively and spontaneously to unforeseen situations [43] and emotional commitment refers to their affective connection to the organization [40]. Both constructs are imperative in dynamical evolving digital settings where unforeseen issues arise and employees' readiness to remain engaged in crucial. Current literature focuses on digital leadership and its influence on organizations [52] and business processes [74]. However, less attention has been paid to psychological processes through which digital leadership enhances two types of employee functioning such as improvisational ability and emotional commitment. This loophole limits the theoretical advancements as digital transformations continues to rely on workers who can think adaptively and stay emotionally connected with their organizations.

Flow theory used as an effective framework to explain how digital leadership influences improvisational ability and emotional commitment. Flow at work refers to a temporary but intense state of engrossment and enjoyment when job demands aligns with the employees' skills [6]. Flow at work boosts cognitive flexibility, quick information and positive affect integration factors that enhance improvisational ability and foster greater emotional commitment [66, 68]. Flow at work is a subjective experience with distinct cognitive and emotional consequences [41]. Notably, flow at work differs from intrinsic motivation or emotional commitment. Emotional commitment is a long-lasting organizational attitude and intrinsic motivation is a constant desire to do something [75]. It is necessary to differentiate these constructs to explain why flow at work is positioned as core psychological mechanism in this study.

Based on the flow theory, we proposed that flow at work mediates between digital leadership and improvisational ability, as well as emotional commitment. This approach is consistent with theoretical assumptions that flow at work emerges only when leadership establishes a skill-challenge balance [64] and provides supportive feedback [38]. Flow at work initiates cognitive and emotional processes underlying improvisation ability and emotional commitment. This theoretical mechanism fills a critical research gap. The available literature barely examines digital leadership' potential to foster these two types of employee functioning.

The extent to which flow at work leads to improvisation ability and emotional commitment depends on the interpersonal climate. Psychological safety reflects shared beliefs that individuals can ask questions, share ideas and take interpersonal risks without negative consequences [18]. When psychological safety is high, employees are more likely to apply the cognitive flexibility [48] and positive emotions generated by flow at work. When safety is low, employees may hesitate to act, limiting flow's benefits. Therefore, we position psychological safety as a moderator that strengthens or weakens the effect of flow at work on improvisational ability and emotional commitment. This boundary role is theoretically appropriate because psychological safety does not directly create flow at work or its' outcomes, but shapes whether employees can translate flow at work into adaptive behavior. Other conceptual roles, such as antecedent or mediator, imply different causal structures that do not align with flow theory.

These theoretical arguments form an integrated model where digital leadership serves as an enabling condition, flow at work functions as the core psychological mechanism. Moreover, psychological safety provides the contextual boundary shaping two key dependent variables: improvisational ability and emotional commitment. This study addresses the lack of understanding in prior research regarding pathways through which digital leadership affects unique cognitive and affective employee capabilities to digital transformation. In particular, it offers novel combination of three constructs such as digital leadership, flow at work and psychological safety rarely been studied simultaneously. This study advance understanding of how leader-created digital environments necessitate both cognitive adaptability and affective attachment. The study's significance lies in explaining how and under what conditions digital leadership helps employees to handle unexpected challenges and remain emotionally connected to their organizations. These internal processes are becoming increasingly critical to both theory and practice as firms continue to rely on knowledge workers to make decisions amid technological uncertainty.

In addition to these theoretical contributions, this study employs both PLS-SEM and fsQCA, allowing us to examine linear relationships and alternative causal configurations. This combined analytical approach not only reveals the net effects of digital leadership and flow at work but also identifies multiple pathways through which

various combinations of conditions lead to improvisational ability and emotional commitment. Overall, the study is valuable for explaining how digital leadership influences employees' improvisational ability and emotional commitment, identifying psychological safety as a boundary condition. Moreover, demonstrating the methodological importance of using PLS-SEM and fsQCA to unpack the complex leadership processes in the digital transformation settings.

Theoretical support

Csikszentmihalyi , conceptualized flow as a state of complete engagement and pleasure which occurs when people are involved in challenging tasks that suit their capabilities resulting in full attention and self-motivation. It is a state where time perception is unclear and focus is very sharp. People experience a feeling of inner fulfilment and drive, which improves work performance and creativity. Based on this, Bakker, [6] applied the concept to the organizational setting as flow at work that incorporates three fundamental components namely absorption, enjoyment and intrinsic motivation toward work. Absorption is full concentration on the job at hand, job enjoyment is associated with pleasure and satisfaction of the working process. Moreover, intrinsic motivation focuses on behaviour at work is motivated by interest and enthusiasm in the work itself [13].

This study is founded on flow theory, which suggests that digital leaders encourage employees to enter a flow state by streamlining workflows, facilitating the internalization digital technology, and aligning employees' skills with task demands [72]. Flow theory points out that the external conditions created by leaders do not directly have downstream cognitive or affective effects. Rather these circumstances enable a psychological state to provoke that triggers the mechanism responsible for such outcomes. A flow state fosters a higher level of improvisational ability, allowing creative and flexible responses to uncertainty and change. In addition, experiencing positive feelings and as sense of achievement during flow at work increases employees' emotional commitment and strengthens their connection to the organization [77].

On the basis of these theoretical assumptions, flow at work is placed as the direct psychological process through which digital leadership affects improvisational ability and emotional commitment. Flow theory explicitly explains that people exhibit greater cognitive flexibility, creativity and positive mood only when they enter a flow state [42]. Leadership behaviors by themselves do not trigger these outcomes; rather, they create the preconditions that make flow possible, such as balanced challenges, clear goals and constructive feedback. This logic provides the justification for conceptualizing flow at work as a mediator: as leadership shapes the conditions for flow at work, which in turn generates the cognitive and emotional processes that lead to these two outcomes. In other words, digital leadership affects improvisational ability and emotional commitment primarily when employees enter a flow state, supporting the view that adaptive responses arise from experience rather than external inputs alone. This positioning rules out direct effects at the theoretical level and supports the mediation structure of the model. Hence, this study constructs the theoretical framework of "digital leadership →flow at work → improvisation ability/emotional commitment and psychological safety" proposes the following research framework shown in Fig. 1.

Hypothesis development

Digital leadership and flow at work

Flow is a psychological state observable across many human activities. Csikszentmihalyi, [13] describes flow as complete immersion and intrinsic enjoyment arising when an individual's skills match task challenges. In work settings, flow at work represents a short-term optimal experience characterized by absorption, satisfaction and intrinsic motivation [66]. Achieving this state requires a balance between challenge and ability; when employees perceive alignment between their capabilities and job demands, they experience control, focus and a sense of accomplishment [50].

Traditional leadership styles such as transactional and transformational leadership emphasize control, structure and relational support [15]. This tends to ignore task-level processes including real-time feedback, digital tools and work redesign which assist employees to enter flow. However, digital leadership incorporates technology to streamline the work processes, offer immediate feedback and match the skills and complexity of the tasks [53], thereby contributing to flow experiences [16]. Digitalization also transforms communication and coordination, making the more flexible and providing better access to information [8]. Digital leaders help employees internalize digital tools, boosting their confidence and enabling a stable balance of skills and challenges [10]. Additionally, since automation leads to greater career uncertainty, knowledge workers invest in skill development, a process that helps them to reach a level where they can fully focus on and enjoy flow at work [9]. Thus:

> *H1*. Digital leadership has a positive impact on flow at work.

Flow at work and improvisational ability

Flow at work refers to a state where employees are completely engaged, self-motivated and working at an ideal challenge-skill ratio [6]. Improvisational ability is defined as a spontaneous and practicable response to unforeseen

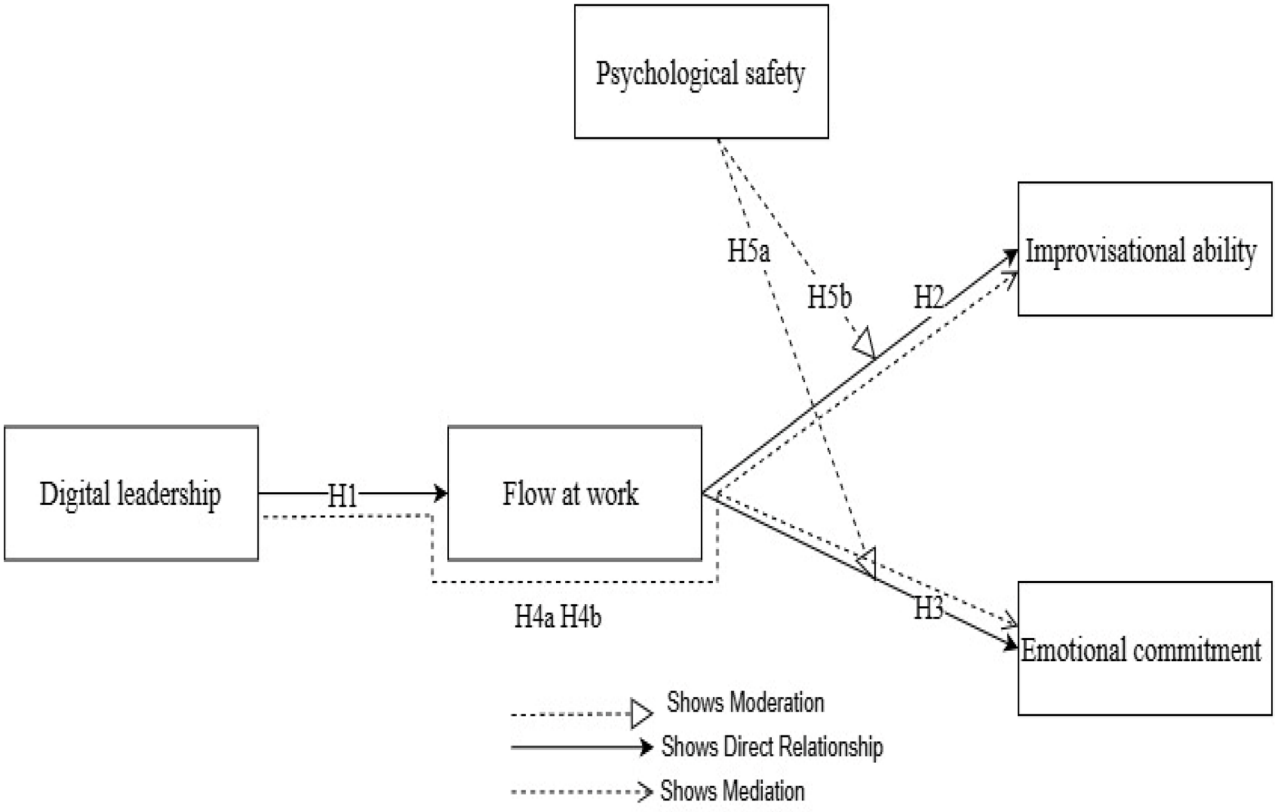

Fig. 1 Study framework

circumstances when routines are inapplicable [69]. Flow at work enhances cognitive flexibility, faster recognition of cues and better integration of multifarious knowledge that would be required in improvisational abilities [49]. Studies indicate that flow at work boosts divergent and convergent thinking, enabling the development of novel and contextually appropriate solutions [4, 65]. The enjoyment derived from flow at work promotes the experimentation and strategic risk-taking behaviours that are closely related with the effective improvisation [70]. Flow at work also regulates emotions in a stressful workplace, allowing employees to concentrate amid pressure [37]. In dynamic and knowledge-intensive settings, employees who frequently experience flow are better equipped to handle ambiguity and exploit emergent opportunities [7]. Hence:

> *H2*: Flow at work has positive and significant impact on improvisational ability.

Flow at work and emotional commitment

Emotional commitment refers to an employees' emotional attachment, identification and involvement with an organization [46]. Employees with strong emotional commitment remain in the organization because they typically display greater loyalty and discretionary effort. This level of commitment occurs when employees feel that their work matters, as it relates to their values and supports their personal growth [35]. Affective experiences such as satisfaction, pride and purpose are positive and improve emotional commitment while decreasing turnover intentions [39]. Flow at work experiences that generate pleasure, intrinsic motivation and complete engagement serve to reinforce emotional commitment to the organization [61]. When employees are consistently involved in purposeful and dynamic work, they grow to identify better with organizational aims and become more eager to invest in long term contributions. Emotional commitment strengthens when employees perceive that the organization provides ongoing benefits, resulting in a deeper emotional attachment [45]. Based on this, it is hypothesized that the experience of optimal flow at work increases the chances of employees consolidating their emotional commitment to the organization. Thus:

> *H3*: Flow at work has positive and significant impact on emotional commitment.

The mediating role of flow at work

According to flow theory, individuals who experience flow at work demonstrate heightened concentration

and positive affect [13]. This is consistent with broaden-and-build theory, which proposes that positive emotions expand and strengthen individuals' cognitive and emotional resources [23]. Flow at work enhances both (cognitive) improvisational ability and (affective) emotional commitment among employees. Cognitively, flow at work improves improvisational ability and spontaneously response to a problem [47]. Improvisational ability refers to immediate, non-routine reaction to an issue [58]. Flow at work expands the thinking of the employees and enhances their ability to experiment and act immediately [68]. This increased cognitive bandwidth provides the resource patch working and quick adaptation which are critical to improvisational ability.

Affectively, flow at work fosters joy, satisfaction and engagement, which enhance emotional commitment. Emotional commitment reflects loyalty and a willingness to stay based on positive emotional identification (Rego et al. 2011). Frequent positive experiences at work strengthen belonging and attachment [58]. Flow at work generates a sense of purpose and fulfilment, encouraging employees to internalize organizational goals and build deeper emotional ties [59]. Digital leadership contributes to these outcomes by using intelligent technologies, automation and training to enhance autonomy, clarity and task challenge conditions that trigger flow at work. In turn, flow at work strengthens improvisational ability and emotional commitment [63]. Drawing on the extend literature, the following hypothesis are proposed:

H4a. Flow at work significantly mediates the effect of digital leadership on improvisational ability.
H4b. Flow at work significantly mediates the effect of digital leadership on emotional commitment.

The moderating effect of psychological safety

Psychological safety is a shared belief that a team environment supports interpersonal risk-taking [18]. Flow at work enhances employees' cognitive and motivational readiness to act creatively, but psychological safety determines whether they feel permitted to express improvisational behaviours [68]. Psychological safety fosters experimentation, learning from immediate feedback and sharing incomplete ideas [34], thereby reducing the social cost of trial-and-error and increasing improvisation [51]. Thus, psychological safety amplifies the relationship between flow at work and improvisational ability.

Psychological safety also determines affective internalization of flow at work and transforms these experiences into deeper emotional commitment to the organization. Flow at work generates a positive effect, pride in the performance and intrinsic satisfaction, circumstances that may generate identification and attachment to the organization [14]. However, without psychological safety those enjoyable moments can become overshadowed by fear to speak up, being judged, or fear of not being rewarded for experimenting all of which can reduce the positive effects of flow at work on emotional commitment. In contrast, in climates with high level of psychological safety, the positive experiences resulting from flow are recognized, supported and socialized (via favourable feedback and celebration of small success). This strengthens the affective bond, encourages discretionary effort and loyalty. So, based on the existing literature we proposed that:

H5a: Psychological safety positively moderates the relationship between flow at work and improvisational ability.
H5b: Psychological safety positively moderates the relationship between flow at work and emotional commitment.

Method

Data collection procedure

Data were collected from full-time employees and their immediate supervisors working in manufacturing and service organizations across multiple regions of Pakistan. Participating organizations were selected purposively to include firms that had experienced notable digital initiatives or environmental dynamism. Organizational contacts were approached and informed about the study objectives and inclusion criteria. Organizations provided lists of teams and employee rosters so that eligible employees and their direct supervisors could be identified.

To reduce common-method bias and ensure independence of data sources, a multi-wave and multi-source design was adopted. The survey was administered in three waves (T1, T2 and T3) with an approximate one-month interval between waves. A one-month lag is consistent with prior methodological recommendations for separating perceptual antecedents, transient psychological states, attitudinal and behavioral indicators [54, 67]. This time frame allows sufficient temporal distance for psychological processes to unfold while minimizing sample attrition. This temporal separation helps to attenuate common method variance and appropriate for capturing possible lagged effects among leaders' behaviours, employee psychological states (flow at work and psychological safety), and attitudinal outcomes (emotional commitment), as well as supervisor-rated improvisational ability.

At T1 (April 2025), employees completed measures of digital leadership and demographic profiles. Total number of distributed questionnaires was 520 at this wave which resulted in 480 received questionnaires. The sample size was 405 and this also represents the final matched sample used in analysis. The demographic

composition of predominantly male (77.7%); majority aged 31–40 years (65.6%); education levels were primarily graduates (64.6%), followed by undergraduates (21.5%) and postgraduates (13.9%); work experience was concentrated in the 6–10 years bracket (51.7%), with 8.1% reporting less than one year of experience. Most respondents were from manufacturing enterprises (78%), with the remainder employed in the service sector (22%) as shown in Table 1.

At T2 (June 2025) one month later, employees provided data on the mediator (flow at work) and the moderator (psychological safety). Total number of the questionnaires distributed was 510, which resulted in 445 completed questionnaires after removing cases with missing identifiers or incomplete questionnaires.

At T3 (August 2025), employees reported their emotional commitment, and immediate supervisors independently rated each employee's improvisational ability. Total 500 questionnaires were distributed. The effective matched set at T3 included 405 employee responses and 405 corresponding supervisor evaluations. Overall, 405 leader–employee dyads were successfully matched across all three waves. Attrition occurred primarily due to employee turnover, leave and unmatched supervisor forms. Missing responses were handled list wise for multi-wave analyses to ensure consistent temporal matching.

Leader and employee surveys were kept separate and matched using unique employee ID codes and name lists supplied by the organizations. This design reduces common method bias by introducing both temporal separation (T1–T3) and source separation (employee versus supervisor response). It also strengthens, but does not overstate, causal inference by ensuring that (a) predictors were measured before mediator, (b) mediator were measured before outcomes and (c) behavioural ratings came from an independent source. Such a design increases confidence in the proposed temporal ordering of effects while acknowledging that causality cannot be definitively proven without experimental manipulation.

Several procedural steps were taken to ensure data quality and confidentiality. Prior to distribution, the research team conducted leader–employee matching by cross-checking names and internal codes to ensure accurate pairing of subordinate responses with their respective supervisors' evaluations. Fixed collection points were established within participating firms to allow employees to return completed questionnaires independently and without researcher-mediated discussions. Participation was voluntary and confidentiality of responses was guaranteed in writing. After data collection, questionnaires were screened and excluded if they contained excessive missing data on key variables, showed evidence of straight-lining, had implausibly rapid completion times or lacked a matching supervisor response where required.

A modest token of appreciation (a small gift of nominal value) was provided to participants after completion of the final wave. Ethical procedures were followed throughout the study: participants received informed consent statements describing the aims, the voluntary nature and confidentiality provisions. Following standard recommendations for sample adequacy in multivariate modelling, the final employed sample of 405 exceeded the commonly cited threshold of 5–10 times the number of observed indicators [27], ensuring sufficient statistical power for the planned Partial Least Squares Structural Equation Modelling (PLS-SEM) analyses [76]. Moreover, sample size was also appropriate for fuzzy-set Qualitative Comparative Analysis (fsQCA), which focuses on configurational pathways rather than linear effects. The use of both PLS-SEM and fsQCA enables a comprehensive analysis that captures net effects and identifies multiple equifinal combinations leading to high improvisational ability and emotional commitment.

Table 1 Demographic Profile

	N	% age		*N*	%age
Gender			Experience		
Male	315	77.7	Less than 1 year	33	8.1
Female	90	22.3	1–5 Years	78	19.3
Age			6–10 Years	209	51.7
20–30	55	13.5	11–15 Years	48	11.8
31–40	265	65.6	16–20 Years	37	9.1
41–50	85	20.9	Nature of Enterprises		
Qualification			Manufacturing	316	78.0
Undergraduates	87	21.5	Services	89	22.0
Graduates	262	64.6			
Post Graduates	56	13.9			

N = 405

Measurement

All the measurements used in our study were adapted from previously published and validated scales. We measured digital leadership using Zeike et al., [74] six-item scale. The items are "My leader always introduces us to interesting digital tools". We measured flow at work using the measurement scale developed by Bakker [6], including the three dimensions of absorption, work enjoyment and intrinsic work motivations comprise of thirteen items, including "When I am working, I think about nothing else". We used seven items from Edmondson [18] to measure psychological safety. Sample items include "Working with members of this team, my unique skills and talents are valued and utilized". We measured improvisational ability using a three item scale from [68]. Items included "In designing this innovation, the employee is very adept at spontaneously dealing with unexpected problems". Yao et al., [73] developed measure to evaluate emotional commitment, consisted of four items, namely "I am very happy to work in this team".

Results

Model measurement

This research model, based on 33 items across five constructs, demonstrates strong reliability as indicated by Cronbach's alpha [28]. As shown in Table 2, each of the items has strong reliability with the Cronbach alpha (α) being greater than 0.7. Moreover, the composite

Table 2 Model measurement

Variables		Item Loading	AVE	CR	α
Digital leadership					
	DL1	0.718	0.652	0.898	0.893
	DL2	0.798			
	DL3	0.788			
	DL4	0.850			
	DL5	0.843			
	DL6	0.840			
Flow at work					
	FAW1	0.728	0.517	0.924	0.921
	FAW2	0.631			
	FAW3	0.747			
	FAW4	0.774			
	FAW5	0.713			
	FAW6	0.756			
	FAW7	0.691			
	FAW8	0.840			
	FAW9	0.747			
	FAW10	0.643			
	FAW11	0.707			
	FAW12	0.703			
	FAW13	0.635			
Emotional commitment					
	EC1	0.841	0.687	0.853	0.848
	EC2	0.833			
	EC3	0.800			
	EC4	0.840			
Improvisational ability					
	IA1	0.849	0.687	0.784	0.768
	IA2	0.903			
	IA3	0.725			
Psychological safety					
	PS1	0.760	0.605	0.896	0.892
	PS2	0.837			
	PS3	0.783			
	PS4	0.806			
	PS5	0.781			
	PS6	0.774			
	PS7	0.699			

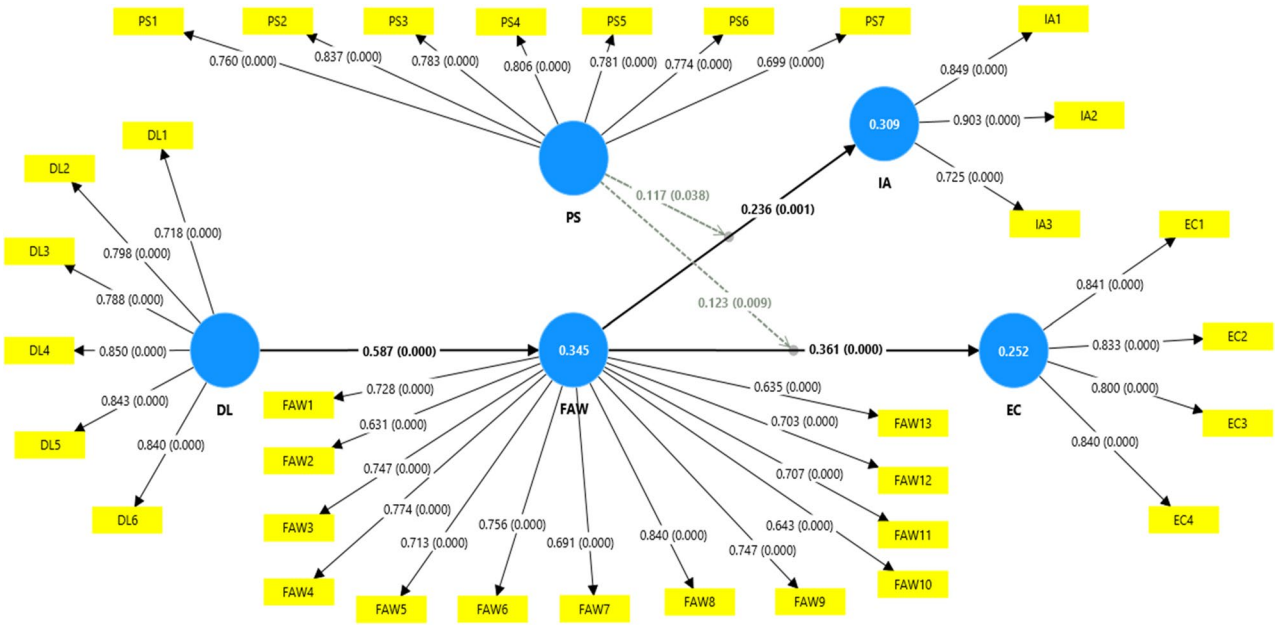

Fig. 2 Measurement model

Table 3 Discriminant validity through HTMT

	DL	EC	FAW	IA	PS
DL					
EC	0.237				
FAW	0.642	0.525			
IA	0.314	0.450	0.528		
PS	0.543	0.393	0.573	0.573	

DL Digital leadership, *FAW* Flow at work, *IA* Improvisational ability, *EC* Emotional commitment and *PS* Psychological safety

reliability (CR) is 0.784 to 0.924, which exceeds the suggested threshold 0.70, which validates the indicative reliability of all factor loadings employed in the current study. Average variance extracted (AVE) was used to test convergent validity according to Fornell and Larcker [20] with an excess of above-recommended 0.50 [26]. Additionally, all items loadings are above the 0.6 cut-off, which is the set threshold [30]. Furthermore, all item loadings surpass the 0.6 cut-off, meeting the established threshold. For digital leadership, item loadings range from 0.718 to 0.850 (AVE = 0.652; CR = 0.898; α = 0.893), while flow at work demonstrates loadings between 0.631 and 0.840 across its 13 indicators (AVE = 0.517; CR = 0.924; α = 0.921). Emotional commitment shows strong factor loadings between 0.800 and 0.841 (AVE = 0.687; CR = 0.853; α = 0.848), and improvisational ability exhibits item loadings ranging from 0.725 to 0.903 (AVE = 0.687; CR = 0.784; α = 0.768). Psychological safety also shows a high item reliability with loadings between 0.699 and 0.837 (AVE = 0.605; CR = 0.896; a = 0.892). Such loading patterns suggest that every construct is within acceptable reliability measures and has reasonable convergence to the desired latent variables.

According to Hair et al. [29], it is considered satisfactory for the Cronbach's alpha value of all constructs to be higher than 0.70. Figure 2 shows that all item loadings exceed the acceptable threshold, confirming strong measurement reliability.

Discriminant validity through HTMT

The HTMT technique presented by [30], was employed in this study. The discriminant validity of the HTMT approach was evaluated using two different techniques. Initially, the threshold value was determined through the HTMT approach. The cut-off value was initially established based on HTMT. Value that exceed the HTMT threshold indicated a lack of discriminant validity. As the correlation approached 1, the precise threshold value of HTMT became a topic of debate. The HTMT values range from 0.237 to 0.642. These values are well below the recommended thresholds of 0.85 [57] as shown in Table 3.

Discriminant validity through Fornell-Larcker criterion

Fornell and Larcker [21] approach was used to examined the discriminant validity. The value of variance of

Table 4 Discriminant validity through Fornell-Larcker criterion

	DL	EC	FAW	IA	PS
DL	0.808				
EC	0.210	***0.829***			
FAW	0.587	0.471	***0.791***		
IA	0.259	0.365	0.456	***0.829***	
PS	0.490	0.351	0.532	0.492	***0.778***

DL Digital leadership, *FAW* Flow at work, *IA* Improvisational ability, *EC* Emotional commitment and *PS* Psychological safety

Table 5 Predictive accuracy and relevance of the model

Construct	(R^2)	(Q^2)
EC	0.252	0.106
FAW	0.345	0.337
IA	0.309	0.211

FAW Flow at work, *IA* Improvisational ability, *EC* Emotional commitment.

the construct in the model with its indicators must be relatively higher than the variance of the other latent constructs. Additionally, square root values of average variance extracted should be higher than maximum shared variance and average shared value, which also confirms discriminant validity. The values of correlation between independent variables' pair must be lower than 0.9 value [24]. The typical values ranges from 0.210 to 0.829. The results are adequately complying with the set criteria, presented in Table 4.

Predictive accuracy and relevance of the model

Table 5 presents the predictive accuracy (R^2) and predictive relevance (Q^2) for the three endogenous constructs: emotional commitment, flow at work, and improvisational ability. The R^2 values indicate that the model explains 34.5% of the variance in flow at work, 30.9% of the variance in improvisational ability, and 25.2% of the variance in emotional commitment, demonstrating moderate predictive accuracy for flow at work and improvisational ability, and a relatively lower yet acceptable level of predictive accuracy for emotional commitment [28]. The Q^2 values further confirm predictive relevance, with flow at work showing strong predictive relevance (Q^2 = 0.337), improvisational ability showing moderate predictive relevance (Q^2 = 0.211), and emotional commitment showing lower but still meaningful predictive relevance (Q^2 = 0.106). These findings indicate that the model most powerfully predicts flow at work, then improvisational ability and, less significantly, emotional commitment, which aligns with the recommendations of [31]. Overall, the results indicate that the structural model has sufficient predictive accuracy and predictive relevance among the primary endogenous constructs.

Hypothesis testing

This study explores the importance of relationships using bootstrapping at 5,000 with replacement sample [5]. The statistical results indicate in Table 6 that digital leadership has a significant association with flow at work (b = 0.587, t = 12.624, and p = 0.000), which supports H1. Furthermore, flow at work significantly influences improvisational ability (β = 0.236, t = 3.386, p = 0.001), confirming H2. Additionally, the findings reveal a positive relationship between flow at work and emotional commitment (β = 0.361, t = 5.702, p = 0.000), supporting H3. Consequently, the direct hypotheses H1, H2, H3 were all accepted.

The mediation analysis corroborates both hypotheses. H4a illustrates that flow at work significantly mediates the connection between digital leadership and improvisational ability (β = 0.139, t = 3.383, p = 0.001), whereas H4b indicates that flow at work also significantly mediates between digital leadership and emotional commitment (β = 0.212, t = 5.433, p = 0.000). Regarding

Table 6 Hypothesis testing

Hypothesis	Relationship among constructs	β	Mean	SD	T values	*p*-values	Remarks
Direct Effect							
HI	DL◊FAW	0.587	0.589	0.046	12.624	0.000	Supported
H2	FAW◊IA	0.236	0.237	0.070	3.386	0.001	Supported
H3	FAW◊EC	0.361	0.361	0.063	5.702	0.000	Supported
Mediation Effect							
H4a	DL◊FAW◊IA	0.139	0.139	0.041	3.383	0.001	Supported
H4b	DL◊FAW◊EC	0.212	0.213	0.039	5.433	0.000	Supported
Moderation Effect							
H5a	PS*FAW◊IA	0.117	0.114	0.056	2.075	0.038	Supported
H5b	PS*FAW◊EC	0.123	0.124	0.048	2.595	0.009	Supported

DL Digital leadership, *FAW* Flow at work, *IA* Improvisational ability, *EC* Emotional commitment and *PS* Psychological safety

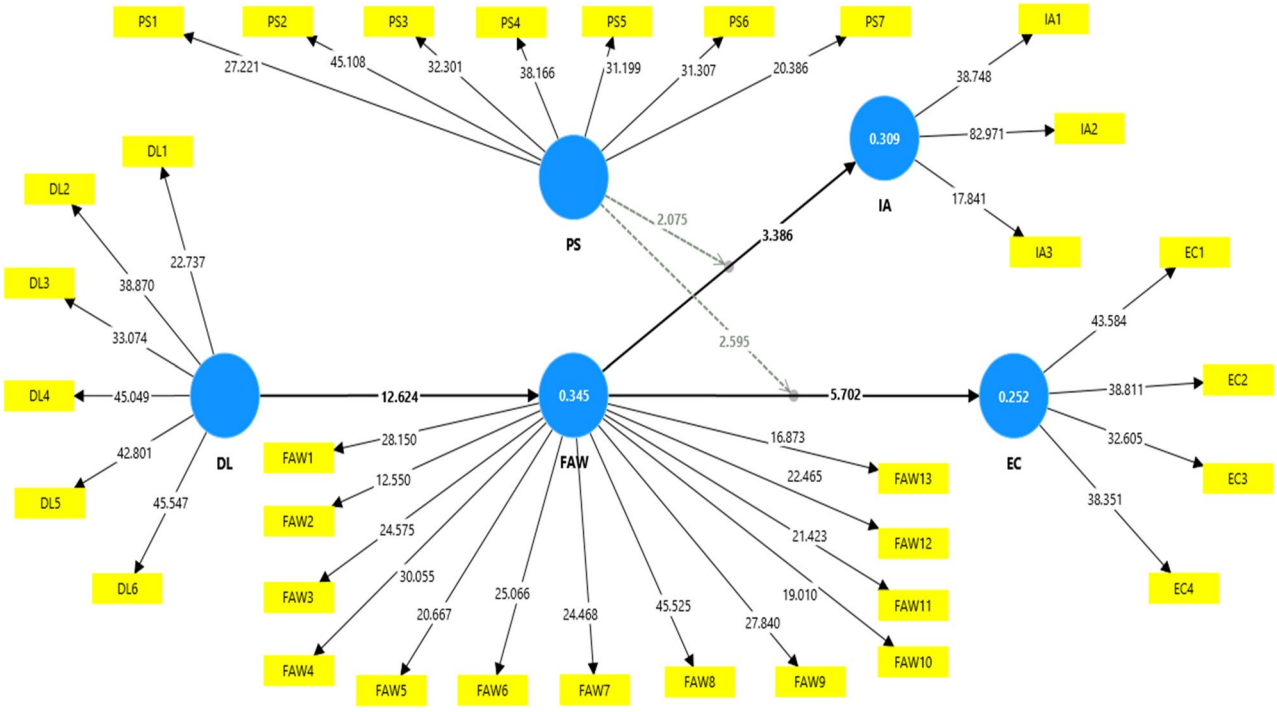

Fig. 3 Structural model

Table 7 Calibrations and descriptive statistics of the research variables

Configurational Construct	Fuzzy Set Calibration							
	Fully in	Crossover	Fully out	Mean	S.D	Min	Max	*N* Case
Digital Leadership (DL)	27.00	13.00	8.00	15.43	5.95	6.00	30.00	405
Psychological Safety (PS)	18.00	9.00	4.00	18.11	7.10	7.00	35.00	405
Flow at Work (FAW)	30.00	16.00	8.00	33.40	11.56	14.00	64.00	405
Improvisational Ability (IA)	58.00	30.00	18.00	7.84	3.07	3.00	15.00	405
Emotional Commitment (EC)	14.00	7.00	4.00	10.21	4.22	4.00	20.00	405

Calibration thresholds: [fully in = top quartile 95%, crossover = median 50%, fully out = bottom quartile 5%

moderation, H5a is validated, suggesting that psychology safety positively moderates the influence of flow at work on improvisational ability (β = 0.117, t = 2.075, $p = 0.038$). Additionally, H5b depicted that psychology safety positively moderate the relationship between flow at work and emotional commitment (β = 0.123 t = 2.595, $p = 0.009$) (Fig. 3).

Data calibration

The first step in integrating fsQCA into this model involves calibrating the PLS-SEM data into fuzzy sets [71]. Moreover, the calibration of the exogenous and endogenous causal constructs must be primarily conducted into fuzzy sets with scores that range from 0 to 1, where a complete set membership is denoted by a score of 0.95, and the crossover value is indicated by 0.50, reflecting no membership of 0.05 [56]. In this study, calibration followed the direct method, applying the 95th, 50th, and 5th percentiles as thresholds for full membership, crossover, and full non-membership respectively, consistent with Crespo et al., [12]. In fsQCA, the breakpoints for all causal conditions and the outcome indicators were calibrated. Table 7 illustrates the descriptive statistics and fuzzy set calibrations for all the exogenous and endogenous causal constructs.

Necessary condition

Once the data is calibrated into fuzzy sets the next step is to perform necessary condition analysis (NCA). An analysis of necessary conditions is one of the most crucial assessments of fsQCA [11]. Before delving into causal configurations, it is always wise to study prerequisites, since it is a sound approach to evaluate a truth table of each outcome [36]. Subsequently, we noted that NCA must focus on the presence or absence of causal conditions that lead to the presence or absence of equivalent implications [60]. Following Ragin and Fiss [55], a consistency score ≥ 0.90 indicates a strictly necessary condition, while scores between 0.80 and 0.90 reflect "almost always necessary" conditions; coverage values above 0.75

Table 8 Necessary Conditions Analysis (IA and EC as Outcome)

Condition	Improvisational Ability (IA)		Emotional Commitment (EC)	
	Consistency	Coverage	Consistency	Coverage
DL	0.738718	0.683673	0.732698	0.725534
~DL	0.569692	0.571745	0.599980	0.644262
PS	0.806103	0.745718	0.777032	0.769107
~PS	0.516154	0.518254	0.544670	0.585141
FAW	0.780615	0.765617	0.760736	0.798311
~FAW	0.554513	0.524444	0.571127	0.577942

~ indicates the absence of a condition. *DL* Digital Leadership, *PS* Psychological Safety and *FAW* flow at work

Table 9 Configurations of achieving high IA and EC

Configuration	Raw Coverage	Unique Coverage	Consistency
Improvisational Ability (IA)			
PS*FAW	0.69277	0.68972	0.834661
Emotional Commitment (EC)			
PS	0.777032	0.105541	0.769107
FAW	0.760736	0.0892448	0.798311

DL Digital Leadership, *PS* Psychological Safety and *FAW* flow at work

signal substantial empirical relevance. All the causal conditions of Table 8 are considered in this analysis.

In the case of improvisational ability (IA) as the outcome, none of the conditions reached the 0.90 consistency threshold, which means that there is no single factor that is absolutely required to achieve high IA. Nonetheless, the consistency of psychological safety (PS) was the highest at 0.806 and the coverage was 0.745 and flow at work (FAW) had a consistency of 0.780 and coverage of 0.765. The two are highly relevant as facilitative enablers to IA. Digital leadership (DL) had a moderate level of consistency of 0.738 with the coverage of 0.683, which implies its significant role, but not as a mandatory requirement. The lack of these conditions (~PS, ~FAW, ~DL) demonstrated significantly lower consistencies (between 0.516 and 0.569), meaning that their lack does not imply the lack of high IA.

In the case of emotional commitment (EC) as the outcome, PS once again reported the highest consistency (0.777) with a coverage of 0.769, then FAW (consistency = 0.760, coverage = 0.798) and DL (consistency = 0.732, coverage = 0.725). These findings indicate PS and FAW as specifically helpful conditions in support of EC. Similar to IA, the negation of these conditions displayed lower consistencies (ranging from 0.544 to 0.599), suggesting that the absence of these factors does not necessarily lead to low EC. Overall, while no single condition was strictly necessary for high IA or EC, PS and FAW consistently emerged as the strongest supportive conditions, with DL also contributing meaningfully in both cases as shown in Table 8.

Configurations

The next step in this phase is to find the configurations of all the antecedents that are adequate for achieving high improvisational ability and emotional commitment as the outcomes. In terms of specific solutions, consistency gauges the degree to which a configuration is sufficient for the specific outcome [62]. In this study, we applied a sufficiency consistency threshold of 0.75, which is widely accepted in fsQCA research, and required frequency thresholds suitable for the sample size. For IA, one configuration met the threshold: the combination of psychological safety (PS) and flow at work (FAW) achieved a raw coverage of 0.692, unique coverage of 0.689, and a consistency of 0.834. For EC, two individual conditions met the threshold: PS with a raw coverage of 0.777, unique coverage of 0.105, and a consistency of 0.769; and FAW with a raw coverage of 0.760, unique coverage of 0.089, and a consistency of 0.798.

The configurations use raw coverage to indicate the proportion of the outcome in the sample cases that is substantially explained by each configuration, while unique coverage captures the proportion of the outcome explained exclusively by that configuration [1]. These findings indicate that the combination of PS and FAW is a strong sufficient condition for achieving high IA, whereas for EC, PS and FAW individually serve as sufficient conditions as shown in Table 9.

Results of intermediate solution

Table 10 illustrates the intermediate solutions obtained from fuzzy-set Qualitative Comparative Analysis (fsQCA), revealing three unique configurations (paths) that contribute to enhancing improvisational ability (IA) and emotional commitment (EC). These configurations

Table 10 Results of Intermediate solution for achieving high IA and EC

Configuration	1 (IA)	2 (EC)	3 (EC)
Psychological Safety (PS)	●●	●●	
flow at work (FAW)	●●		●●
Digital Leadership (DL)			
Consistency	0.8347	0.7691	0.7983
Raw Coverage	0.6928	0.7770	0.7607
Unique Coverage	0.6897	0.1055	0.0892
Overall Solution Consistency (IA)	0.8347		
Overall Solution Coverage (IA)	0.6928		
Overall Solution Consistency (EC)	0.7295		
Overall Solution Coverage (EC)	0.8663		

Black Circles (●) indicate the presence of a condition, and Cross circle (⊗) indicate its absence. Large circles indicate core conditions; small ones indicate peripheral conditions. Blank spaces indicate "don't care"

present various combinations of core and peripheral conditions that are either present (●) or absent (⊗), reflecting the principle of equifinality, where multiple causal pathways can produce the same outcome. For IA (Configuration 1), both psychological safety (PS) and flow at work (FAW) appear as core conditions (●●), achieving a consistency of 0.834, raw coverage of 0.692, and unique coverage of 0.689. This highlights the strong combined role of PS and FAW in driving high IA.

For **EC**, two configurations emerged. In Configuration 2, PS appears as a core condition (●●) and achieves a consistency of 0.769, raw coverage of 0.777, and unique coverage of 0.105. In Configuration 3, FAW serves as a core condition (●●) alongside PS, with a consistency of 0.798, raw coverage of 0.760, and unique coverage of 0.089. These results indicate that PS alone or in combination with FAW can sufficiently lead to high EC. The overall solution consistency and coverage values confirm the robustness of these models. For IA, the overall solution consistency is 0.834 with an overall coverage of 0.692. For EC, the overall solution consistency is 0.729 with an overall coverage of 0.866. Collectively, these configurations demonstrate that PS and FAW either independently or in combination are pivotal in fostering high IA and EC.

Discussion

Based on the flow theory, the current study sought to enhance our understanding of the mediating role of flow at work and moderating role of psychological safety in linking digital leadership with cognitive and affective competencies of employees. Therefore, we suggested a study model in which work flow at work intervenes the association of digital leadership with two fundamental employee outcomes; namely improvisational ability (cognitive competence) and emotional commitment (affective competence). Besides directly and indirectly testing the effects through PLS-SEM and fuzzy-set qualitative comparative analysis (fsQCA) was also used in the exploration of multiple causal pathways to achieve high improvisational ability and emotional commitment.

Both PLS-SEM and fsQCA findings demonstrated that digital leadership is an important determinant of flow at work among employees that can largely lead to their substantial improvisational ability and emotional commitment. Flow at work turned out to be a complete mediator, which proves the fact that it is the psychological process through which digital leadership transforms into improvisational ability and emotional commitment. This is in line with flow theory, which states that optimal experience and intrinsic motivation create high performance and long-lasting involvement [32]. Moreover, psychological safety enhanced the positive influences of flow at work on both cognitive and affective outcomes which supports a risk-free and supportive environment.

Notably, the fsQCA findings are supplementary to the SEM findings because they show that high improvisational abilities and emotional commitment are not produced by isolated antecedents but rather specific combinations of conditions. For improvisational ability, psychological safety and flow at work together form a strong and sufficient condition. Employees improvise most effectively when they are deeply engaged in their tasks and feel that their work environment is safe. In the case of emotional commitment, psychological safety and flow at work alone were found to be adequate. These configurational results complement the net-effect rationale of SEM and validate the equifinality principle in the development of employee competence.

Figure 4a indicates that flow at work has positive correlation with emotional commitment, which is enhanced by psychological safety. When psychological safety is low, the slope between flow at work and emotional commitment is positive but modest (slope ≈ 0.476), indicating that increases in flow lead to only slight increases in emotional commitment. However, when psychological safety is high, the slope becomes substantially steeper (slope ≈ 0.968), showing that employees experiencing both high flow at work and high psychological safety report much stronger emotional commitment. This means that psychological safety not only enhances

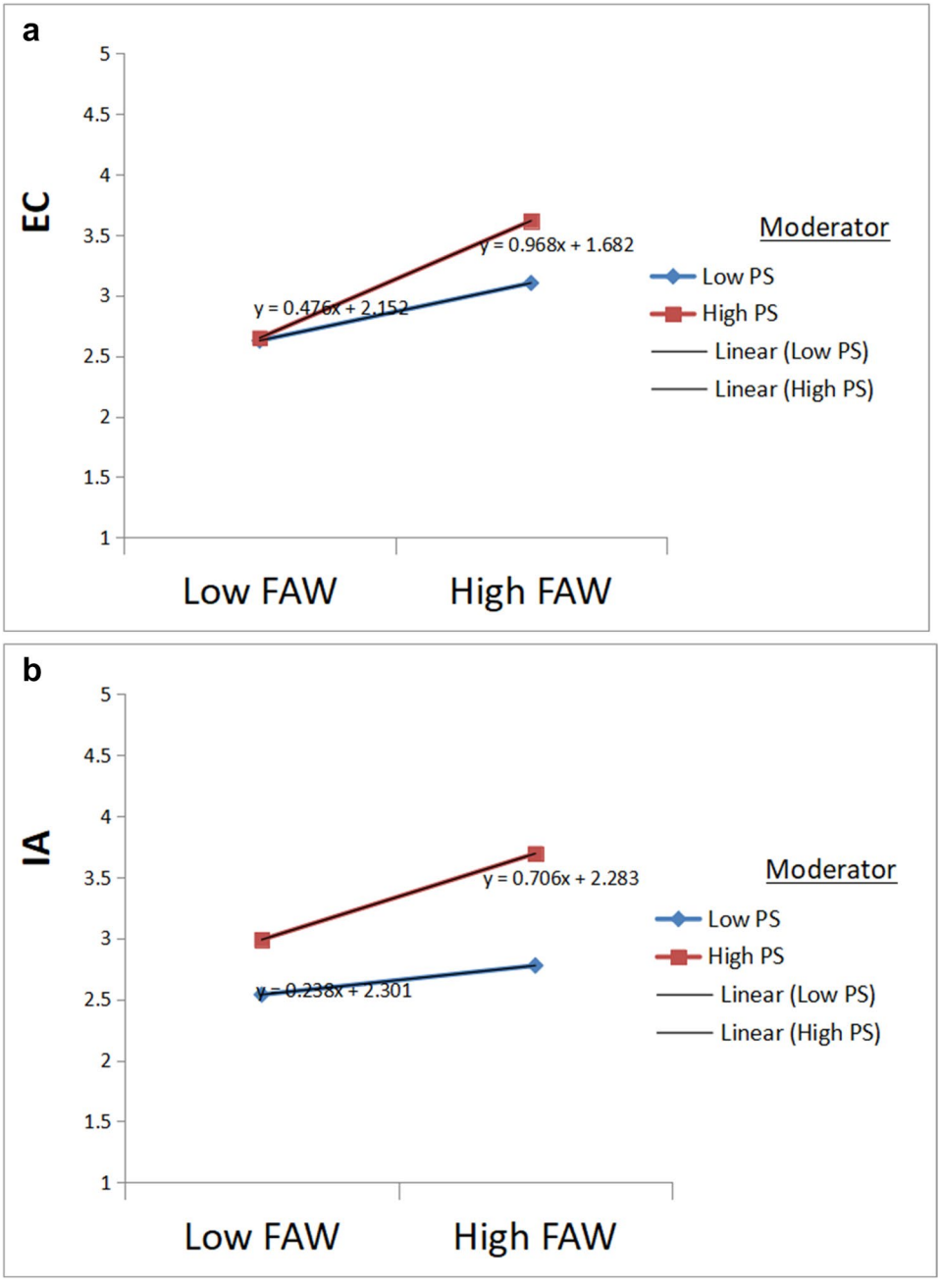

Fig. 4 **a** Interaction effect of psychological safety and flow at work on emotional commitment. Source: Figure by authors. **b** Interaction effect of psychological safety and flow at work on improvisational ability. Source: Figure by authors

but nearly doubles the impact of flow at work on emotional commitment. In other words, employees working in a psychologically safe environment derive significantly greater emotional commitment to the organization from being in a high-flow state compared to those in low-safety environments.

Similarly, Fig. 4b shows that psychological safety strengthens the positive relationship between flow at work and improvisational ability. When psychological safety is low, the slope between flow at work and improvisational ability is positive but shallow (slope ≈ 0.238), indicating that increases in flow at work lead to only minor improvements in improvisational ability. In contrast, when psychological safety is high, the slope becomes substantially steeper (slope ≈ 0.706), demonstrating that employees in psychologically safe environments benefit more from being in a high-flow state. This means that psychological safety not only elevates baseline improvisational ability but also significantly

enhances how strongly flow at work translates into adaptability, spontaneity and creative problem-solving.

The results of the fsQCA further emphasized that combinations of flow at work and psychological safety are a core condition in providing high levels of improvisational ability and emotional commitment, though the single factors of flow at work and psychological safety are insufficient. This highlights the dynamic configuration of employee competence building in virtual environments. Practically, the findings suggest that organizations undergoing digital transformation should promote digital leadership behaviors that foster flow at work. At the same time, they should create psychological safety to support employees' cognitive and emotional development. Overall, this study advances flow theory by demonstrating its applicability in understanding how digital leadership shapes employee outcomes in knowledge-intensive environments. The fsQCA results indicate that Proposition_1 is a sufficient but not necessary condition for high Improvisational Ability (IA). High values of Proposition_1 are often associated with high IA (consistency = 0.8347), but IA can also be high when Proposition_1 is low, as reflected in the lower necessity consistency score (0.6928) as shown in Fig. 5a.

Similarly, Fig. 5b indicates that Proposition_1 is a sufficient but not necessary condition for high emotional commitment. High values of Proposition_1 are often associated with high EC (consistency = 0.8656), but EC can also be high when Proposition_1 is low, as reflected in the lower necessity consistency score (0.67149).

Theoretical implications

First, this study investigates how digital leadership influences employees' improvisational ability and emotional commitment, specifically through the lens of flow at work. This provides a foundation for further research at the individual employee level. Existing literature on digital leadership are primarily focused on examining its antecedent variables and effect on organizational performance and innovativeness [3, 19]. This study also contributes to the body of leadership studies, as well as expands the scope of practical knowledge about digital leadership, by examining it at the level of specific employees. This study underscores the need to promote the cognitive and emotional abilities of the employees. The findings indicate that flow at work suggests more than emotional commitment by digital leadership in promoting the improvisational ability of employees. Therefore, this research highlights the importance of learning the direct effect of digital leadership on individual employee outcomes and gives a more detailed perspective of its overall effect.

Second, this research introduces a new context of the flow theory application and enriches the theory with concrete illustrations of employee cognition and emotions (improvisational ability/emotional commitment). The results show that flow at work is one of the means that allow leaders to activate the potential of employees, enabling them to self-develop and make breakthroughs by supporting their self-actualization awareness and defining their working conditions. In addition, meta-analyses of flow at work reveal that antecedent variables that lead to such a state are primarily divided into three categories, namely job, individual and leadership characteristics. Nonetheless, the leadership literature has primarily focused on traditional forms such as transformational and authentic leadership, with limited research devoted to this area [44].

Third, in contrast to previous literature this study introduces an emerging form of leadership known as digital leadership based on the ongoing trend of digital transformations in enterprises. It also examines the unique influence of digital leadership on the improvisational ability and emotional commitment of individual employees. Furthermore, this study narrows down on the outcome variables of flow at work to include two distinct variables of employee competencies, which further contributes to the theoretical base on differences in flow at work on employees.

Fourth, the paper demonstrates the moderating role of psychological safety in the digital leadership setting and critically analyzes its sophisticated role. Digital technologies are an important factor contributing to the improved efficiency of work and innovation in the era of big data. Psychological safety is the collective belief that the team climate is secure to interpersonal risk-taking [18], which has also gained growing academic interest as a determinant of learning, creativity and performance [22]. Current studies in psychological safety have mostly focused on its advantages and have given minimal consideration to the possibility of boundary conditions at individual levels. We examine the flow of employees at work under digital leadership by exploring the psychological safety as the moderator of improvisational ability and emotional commitment. We discover that the flow at work and these outcomes are dependent on psychological safety to a significant degree, where high levels of psychological safety can moderate the positive impact of digital leadership on emotional commitment through flow at work.

Last, the result can be connected to how psychological safety influences the interpersonal communication patterns and emotional processing of employees [17]. Employees of a high-psychological-safety workplace can feel free to express their concerns, challenge procedures and explore novel opportunities. However, such openness may sometimes dissipate the intensity of immersive work experiences by focusing more on dialogue and feedback and not on a prolonged emotional commitment.

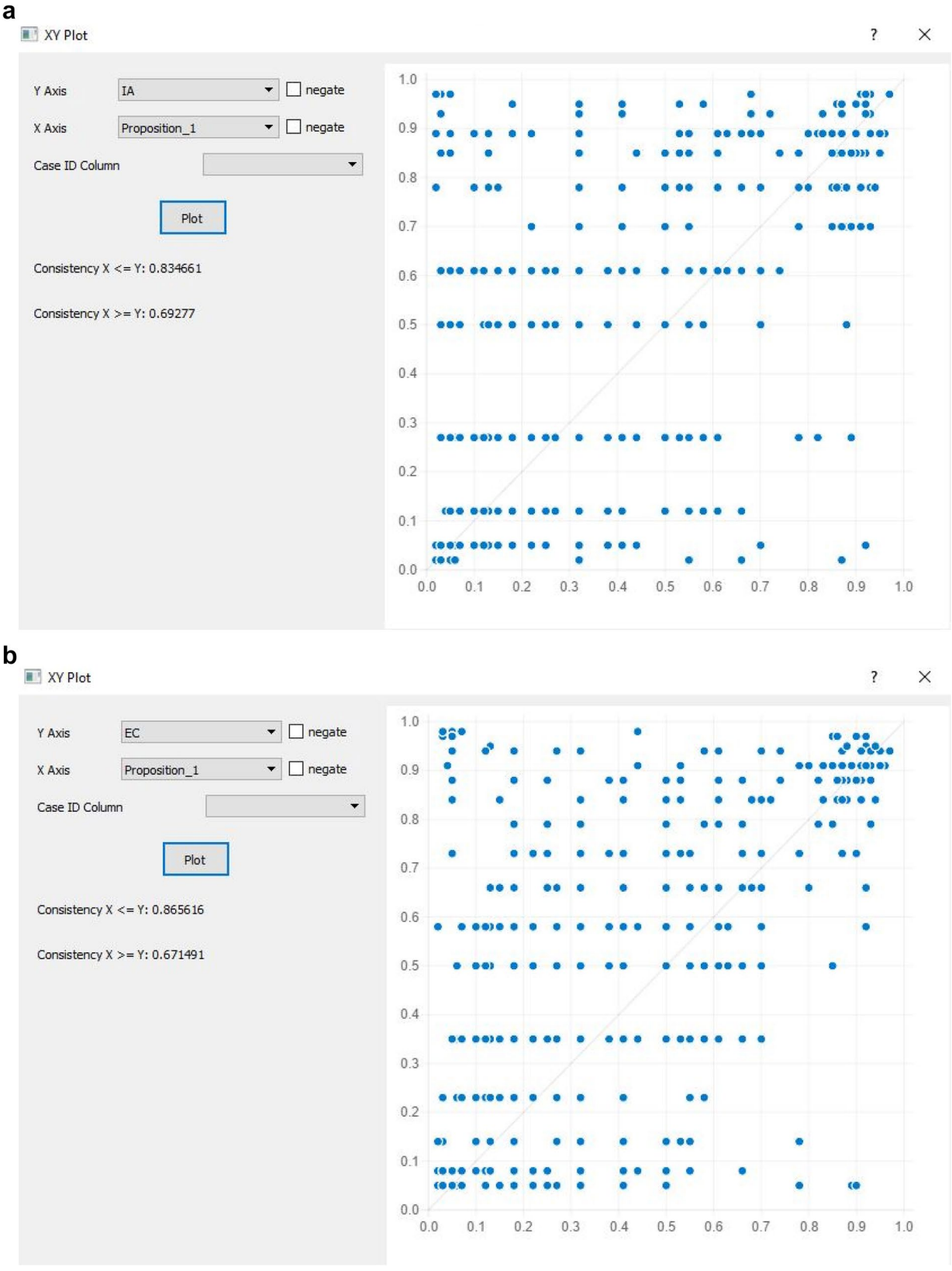

Fig. 5 **a** Plotting a specific proposition (Proposition1; high psychological safety and high flow at work will lead to high Improvisational ability). **b** Plotting a specific proposition (Proposition1; high psychological safety and high flow at work will lead to high emotional commitment)

Practical implications

Our study provides evidence for the positive impacts of digital leadership in enhancing employees' cognitive and affective dimensions by fostering their experience of flow at work in the workplace over time. However, our results show that psychological safety can be excessively high and prevent this process. Although psychological safety usually supports open communication and trust, it may sometimes weaken the link between immersive work experiences and long-term emotional commitment. These insights suggest that digital transformation requires leaders to balance psychological safety with mechanisms that sustain focus, challenge and meaningful engagement.

First, managers should consider developing workplaces that deliberately build flow at work. This includes aligning the employees to the level of task difficulty and establishing precise goals and feedback. The balance of challenge-skill may be actively maintained, and the employees have a higher chance of achieving a deep work state that is linked to a higher capacity of improvisation and emotional commitment. Job rotation, disciplined upskilling schemes and differentiation of task allocation can be used by leaders to balance this in rapidly evolving digital settings.

Second, psychological safety ought to be addressed as a situational resource and it should no longer be perceived as consistently useful. Leaders should also promote interpersonal trust and communication and be responsible in ensuring that employees provide quality performance. Establishes another standard of healthy risk-taking, idea testing and experimenting helps the employees to transform flow experience into adaptive behaviour and not merely to social comfort.

Third, it is essential to balance individual productivity with significant social behaviour. The inclusion of planned face-to-face contact and expressiveness may reinforce relationship-related ties and working relationships, even in digitally mediated settings. At the operational level, the leaders can coordinate face-to-face interaction sessions periodically, progress review and cross-functional knowledge sharing sessions based on milestones in order to build a connection. These initiatives can be supplemented by hybrid onboarding, mentorship programs and recognition (digital appreciation boards, virtual coffee breaks) by HR departments to strengthen social recognition and emotional commitment.

Fourth, the improvisational skill and emotional dedication of employees must be closely observed by the managers as the sign of the effectiveness of the digital transformation processes. There are also the real time feedback systems that leaders can use to know when employees are stimulated and adaptive, by using regular check-ins, reflective debriefs. Such continuous monitoring of the organization allow leaders to redefine task design, team climate and team leadership support in a way that can maintain flow at work over time.

Overall, these practical implications emphasize that digital transformation is not a technological change but also psychological and relational change. Digital leaders foster a cognitively agile and emotionally engaged workforce by deliberately cultivating flow at work and maintaining a balanced level of psychological safety. In addition, they encourage adaptive behaviours and sustain a positive orientation toward the transformation process.

Limitations and future directions

This study has several limitations that open valuable avenues for future research. First, we relied on flow theory to examine how digital leadership influences employees' cognitive and affective outcomes. Future studies could integrate alternative theoretical perspectives such as social exchange theory, job demands-resources theory and affective events theory to capture complementary mechanisms. Second, except for the supervisor-rated improvisational ability, most variables were self-reported and this issue might be subject to common method bias. Even though a time-lagged and multi-source design was adopted in the PLS-SEM framework to counteract this risk, it cannot be eliminated completely. Further studies are needed to include other-rated or objective measures of performance and employ longitudinal designs (e.g. lagged or cross-lagged models) to examine dynamic causal relationships.

Third, while the PLS-SEM results indicated that psychological safety strengthens the positive effect of flow at work on emotional commitment. The fsQCA findings revealed multiple causal configurations leading to high improvisational ability and emotional commitment, the underlying mechanisms remain unclear. Social presence, richness of communication and emotional contagion are some of the variables that may be used to clarify why high psychological safety enhance the emotional gains of flow at work. These are not considered in our dataset and we should study them in further research to determine more about the interaction of flow at work and emotional outcomes under various team conditions. Fourth, given the dual evaluation structure leaders rating employees and employees rating leaders role expectations and social identity tendencies may have influenced responses, potentially affecting rating objectivity. In further studies, third-party ratings behavioural performance scales or social desirability scales of control might be adopted in order to improve validity and strength. Last, since this study was carried out in Pakistan, which is a collectivist society where employees tend to conform to the values and norms of the group, cultural values could have affected the relationships observed. It would be helpful

to replicate this study with other individualistic settings to determine the generalizability of our PLS-SEM and fsQCA results across cultures.

Conclusion

The gradual stimulation of employee competencies is a fundamental driver of sustained growth and innovation in today's rapidly evolving environment. As a key form of contemporary leadership, digital leadership plays a critical role in guiding and motivating employees to enhance their skills. Based on the flow theory, this research explores the impact of digital leadership on employee motivation and the emergence of the bi- dimensional competencies including improvisational ability and emotional commitment. The PLS-SEM findings show that digital leadership positively influences both competencies and flow at work is an important intermediary variable. Furthermore, psychological safety was a positive moderator to strengthen the correlation between flow at work and the improvisational ability and emotional commitment. To support these results, the fsQCA analysis indicates that there are several causal configurations where digital leadership, flow at work, and psychological safety jointly leads to high improvisational ability and emotional commitment, which confirms the principle of equifinality.

Abbreviations

DL	Digital Leadership
PS	Psychological Safety
FAW	Flow at work
IA	Improvisational Ability
EC	Emotional Commitment

Acknowledgments

Not applicable.

Informed consent

This study was conducted in accordance with ethical research guidelines. The survey data were collected from voluntary participants who provided informed consent prior to their involvement. Written consent was obtained from all participants during survey period, which employed the time lagged approach. First round spanned from April 14, 2025 to May 15, 2025, second round spanned from May 18, 2025 to June 19, 2025, and third round spanned from June 20, 2025 to July 21, 2025 using a survey questionnaire. The participants was wholly voluntary, without any risk, and didn't involve any form of compensation.

Authors' contributions

We also confirm that all authors have approved the manuscript for submission. 1. **Author Jianhua Zhang: ** Supervised the research and approved the manuscript before submission. 2. **Author Azmat Yar Khan: ** Writing manuscript, Conceptualization, Data Collection, Editing. 3. **Author Maria Akhtar: ** Manuscript Drafting, Visualization and Validation, Formal Analysis and Software Utilization. 4. **Author Asfand Yar Khan: ** Review and Rewriting of the Manuscript.

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Data availability

The study data are unavailable to the public due to confidentiality agreements with participants. However, anonymized data may be shared with qualified researchers upon reasonable request to the corresponding author, pending ethical approval and adherence to privacy protocols.

Declarations

Ethics approval and consent to participate

This study was reviewed and approved by the *Institutional Review Board (IRB) of School of Management*, *Zhengzhou University*, *China (Approval/Protocol Code: E-Z-2025001; Approval Date: 20 March 2025).* All study procedures, including participant recruitment and data collection in Pakistan, were conducted in accordance with the ethical guidelines approved by this IRB and followed the principles of the *Declaration of Helsinki (1964)* and its later amendments. At the time of data collection, Pakistan did not have a formal institutional ethics review system available for independent researchers; therefore, the Zhengzhou University IRB served as the sole responsible ethical body overseeing the study. All participants were fully informed about the purpose of the research, participation was voluntary, and confidentiality and anonymity were assured. Written informed consent was obtained from all participants prior to each stage of data collection.

Consent for publication

Not applicable.
This study does not include any identifying images or personal details of participants that would require consent for publication.

Competing interests

The authors declare no competing interests.

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